Math 54 — Week 2

Solutions

P1.

Idea. Row reduce $[\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\ \mathbf{b}\,]$; the solution gives the weights.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 2 & 3 & 4 & 12 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 2 & 2 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$x_3 = 3$, $x_2 = 1 + 3 = 4$, $x_1 = 5 - 8 - 3 = -6$. So $\mathbf{b} = -6\mathbf{a}_1 + 4\mathbf{a}_2 + 3\mathbf{a}_3$. Check: $(-6, 0, -12) + (8, 4, 12) + (3, -3, 12) = (5, 1, 12)$.

P2.

Idea. $\mathbf{y}$ is in the span exactly when $[\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{y}\,]$ is consistent.

$$\left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 1 & -5 \\ -2 & 8 & -3 \end{array}\right] \xrightarrow{R_3+2R_1} \left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 1 & -5 \\ 0 & 2 & 2h-3 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{cc|c} 1 & -3 & h \\ 0 & 1 & -5 \\ 0 & 0 & 2h+7 \end{array}\right]$$

Consistent exactly when $2h + 7 = 0$: $h = -7/2$.

P3.

Idea. Columns span $\mathbb{R}^3$ exactly when there is a pivot in every row.

1. A: $R_2 + 4R_1$ gives $(0, 14, 10)$; $R_3 + 3R_1$ gives $(0, 7, 5)$; then $R_3 - \frac{1}{2}R_2$ gives a zero row. Only two pivots: the columns do *not* span $\mathbb{R}^3$.
2. B: $R_3 - 2R_1$ gives $(0, 0, 1)$. Echelon form
$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}:$$
three pivots, one in every row. The columns span $\mathbb{R}^3$.

P4. Definition:

$$5\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - 2\begin{bmatrix} -1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10+2 \\ 0-6 \\ 5-8 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \\ -3 \end{bmatrix}.$$

Row-vector rule: $2 \cdot 5 + (-1)(-2) = 12$; $0 \cdot 5 + 3(-2) = -6$; $1 \cdot 5 + 4(-2) = -3$. Same answer.

P5.

Idea. Reduce, solve for basic variables, split by free variable.

$$\begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \\ 2 & 7 & -10 & 19 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 9 & -8 \\ 0 & 1 & -4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(Steps: $R_3 - 2R_1$ gives $(0, 1, -4, 5)$, then $R_3 - R_2$ gives a zero row; finally $R_1 - 3R_2$.) $x_1 = -9x_3 + 8x_4$, $x_2 = 4x_3 - 5x_4$, x_3, x_4 free:

$$\mathbf{x} = x_3 \begin{bmatrix} -9 \\ 4 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 8 \\ -5 \\ 0 \\ 1 \end{bmatrix}.$$

P6.

Idea. Particular solution plus the homogeneous solutions.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 3 & 7 & 0 & 5 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & -7 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 = -3 + 7x_3$, $x_2 = 2 - 3x_3$:

$$\mathbf{x} = \begin{bmatrix} -3 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -3 \\ 1 \end{bmatrix}.$$

A line through $(-3, 2, 0)$ parallel to $(7, -3, 1)$; the solution set of the homogeneous system is the parallel line through $\mathbf{0}$. Check: $A(-3, 2, 0) = (1, 4, 5)$ and $A(7, -3, 1) = (0, 0, 0)$.

P7.

1. (a) False. A 3×2 matrix has at most 2 pivots, so not one in every row.
2. (b) True. By the theorem on solution sets, the solutions are $\mathbf{p} + \mathbf{v}_h$ with $\mathbf{v}_h$ a solution of $A\mathbf{x} = \mathbf{0}$; the only such $\mathbf{v}_h$ is $\mathbf{0}$.
3. (c) True. At most 3 pivots among 5 columns, so at least 2 free variables; the homogeneous system is consistent, so it has nontrivial solutions.

P8.

Idea. One equation, one basic variable x_1, two free variables.

$x_1 = 4 - 2x_2 + 3x_3$:

$$\mathbf{x} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}.$$

The homogeneous equation has solution set $\text{Span}\{(-2, 1, 0), (3, 0, 1)\}$, a plane through $\mathbf{0}$. The given equation's solution set is that plane shifted by $(4, 0, 0)$: a parallel plane not through $\mathbf{0}$.